DAEN/ISEN 427 In class/homework assignment

Due: Tue Oct 7 9:30 AM

You may use a calculator or short Python snippets. If using Python, include both code and numeric output.

Q1. Over 3 days you observed nearby cars: $x = [3, 4, 1]$. Assume counts are i.i.d. Poisson with rate λ.

a) Compute $\hat{\lambda}_{\text{MLE}}$ and $P(X > 0 \mid \hat{\lambda})$ for a new day. $P(X > 0 \mid \lambda) = 1 - e^{-\lambda}$

Answer:

b) Put a prior $\lambda \sim \text{Gamma}(\alpha = 2, \beta = 2)$ (where β is the *rate*). Compute posterior parameters α', β' for $\lambda \mid x \sim \text{Gamma}(\alpha', \beta')$.

Answer:

c) For a new day X_{new}, use the Gamma–Poisson predictive (Negative Binomial). Show $P(X_{\text{new}} = 0 \mid x) = \left(\frac{\beta'}{\beta'+1}\right)^{\alpha'}$ and compute $P(X_{\text{new}} > 0 \mid x)$.

Answer:

Q2. A coin has head-probability $p \sim \text{Beta}(\alpha, \beta)$. You flip $N = 10$ times and see $y = 7$ heads.

a) With uniform prior $\text{Beta}(1, 1)$, find posterior α', β'.

b) Compute posterior mean $\mu_{\text{post}} = \frac{\alpha'}{\alpha' + \beta'}$.

c) Compute posterior variance

$$\sigma^2_{\text{post}} = \frac{\alpha' \beta'}{(\alpha' + \beta')^2 (\alpha' + \beta' + 1)}.$$

d) Repeat (a)–(c) with informative prior $\text{Beta}(4, 2)$. Compare.

Answer:

Q3. For a Beta$(1,1)$ prior on p, the posterior mean after N observations is

$$\mu_{\text{post},N} = \frac{y+1}{N+2}.$$

a) Show that after observing $x_{N+1} \in \{0,1\}$,

$$\mu_{\text{post},N+1} = \mu_{\text{post},N} + \frac{1}{N+3}\big(x_{N+1} - \mu_{\text{post},N}\big).$$

b) Starting from no data, process $x = [1,0,1,1,0]$. Compute μ_{post} after each.

Answer:

Q4. With flat prior on s, two independent measurements: $x_1 \sim \mathcal{N}(s, \sigma_1^2), \qquad x_2 \sim \mathcal{N}(s, \sigma_2^2).$

a) Derive or state:

$$\mu_{\text{post}} = \frac{\frac{x_1}{\sigma_1^2} + \frac{x_2}{\sigma_2^2}}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}}, \qquad \sigma_{\text{post}}^2 = \frac{1}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}}.$$

b) Numerical practice: $x_1 = 10, \sigma_1 = 2$; $x_2 = 14, \sigma_2 = 4.$

Answer: